

Crustal velocity structure around the eastern Himalayan syntaxis: Implications for the nucleation mechanism of the 2017 M_s 6.9 Mainling earthquake and regional tectonics

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ABSTRACT

A magnitude (M_s) 6.9 earthquake occurred on November 18, 2017, in Mainling, southeastern Tibet, which is the largest earthquake to occur in or around the eastern Himalayan syntaxis within the last 50 years. To further understand the nucleation mechanism of this earthquake, the local seismicity distribution, and the regional tectonics, the detailed crustal velocity structure is obtained by direct ambient noise tomography using the Rayleigh wave phase velocity data ($T = 5 \sim 40$ s) in this area. The spatial correlation of the velocity model with the distribution of the seismicity reveals that the nucleation mechanism of the 2017 M_s 6.9 Mainling earthquake is closely related to the local stress condition and velocity structure and to the aqueous fluid and geothermal states. Our velocity model is extensively compared with previous magnetotelluric inversion results and reveals the distribution and connectivity of the crustal low velocity zones (LVZs). The general patterns of the velocity model in the middle crust show LVZs in the northwest and high velocity zones in the southeast that are truncated by major faults. We also find a relatively low-velocity anomaly in the upper crust near the Namche Barwa massif; this can be explained by the conceptual 'tectonic aneurysms' model, which emphasizes the coupling effects of climate, erosion, and tectonics. To reconcile our velocity model with previous observations, we propose a tentatively integrated geodynamic model to explain the regional seismicity and stress condition.

1. Introduction

On November 18, 2017, a magnitude (M_s) 6.9 earthquake occurred in Mainling County, Nyingchi Region, Xizang Autonomous Region, China (Zhang et al., 2018). Preliminary results show that this earthquake occurred at the location 29.75°N , 95.02°E in southeastern Tibet and that it had a source depth at roughly 10 km (Fig. 1). Detailed source parameters of the 2017 Mainling earthquake have been studied by several research groups using different datasets (Table 1). Although these datasets contain different details, the results share the same features, indicating that this earthquake was thrust-type and occurred in the upper crust. The preliminary finite fault inversion result shows that the rupture direction is to the east-south-east (<http://www.cea-igp.ac.cn/tpxw/276003.html>). Recent seismic relocation studies have shown that the earthquake occurred on the Xixingla Fault (Wei et al., 2018; Yin et al., 2018), which is in the west-north-west direction and is approximately parallel to the Jiali Fault.

The nucleation mechanism of large earthquakes is closely related to

the local velocity structure and stress condition. The collision of the Indian and Eurasian plates since approximately 55 Ma has introduced the general north-south (N-S) shortening and east-west (E-W) extension in the Tibetan Plateau (TP), resulting in the dramatic uplift and thickening of the underlying crust (e.g., Yin and Harrison, 2000; Li et al., 2015). The 2017 Mainling earthquake occurred in the eastern Himalayan syntaxis (EHS) area, also called the Namche Barwa syntaxis, which is in the southeastern corner of the TP (Fig. 1). The EHS is part of the Himalayan Thrust Belt and has acted as an indenter squeezing into the Lhasa Block since 40 Ma (e.g., Zurek, 2008; Xu et al., 2013), forming a large antiform that plunges eastward beneath the adjacent Lhasa Unit (Geng et al., 2005; Burchfiel and Chen, 2012) (Fig. 1). The EHS suffers strong stress concentration and deformation localization, rapid rock uplift and exhumation, a high erosion rate (5–17 mm/yr), powerful fluvial activity (~ 200 kW/m), heavy river incision, and intense rock metamorphism (e.g., Zeitler, 2001; Finnegan et al., 2008; Enkelmann et al., 2011; Koons et al., 2013; Gupta et al., 2015; Lin et al., 2017). In addition, a strong elevation of isotherms and high near-

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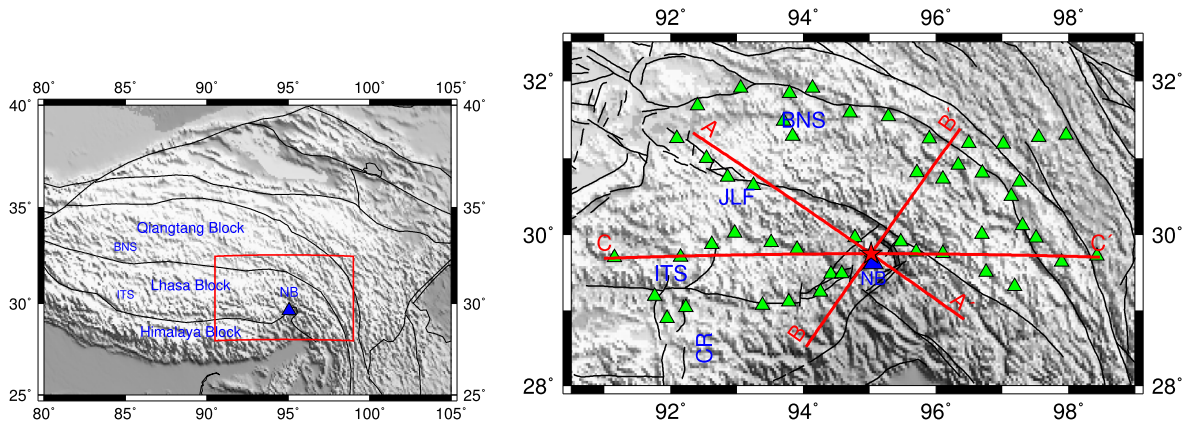


Fig. 1. Tectonic settings and distribution of stations. Red rectangle in the left figure denotes the study area, which is enlarged on the right. Green triangles denote the station locations. The blue triangle denotes the location of the Namche Barwa peak. The red star denotes the location of the Mainling earthquake. Red lines in the figure on the right denote the cross sections crossing the location of Mainling earthquake, which is discussed in detail in Fig. 8 and Fig. 9. ITS, Indus-Tsangpo suture; BNS, Bangong-Nujiang Suture; JLF, Jiali Fault; NB, Namche Barwa peak; CR, Comei rift.

Table 1

Source parameters of the Mainling earthquake determined by different research groups. Only parameters of one nodal plane are listed. CENC, <http://www.ceic.ac.cn>; USGS, <https://earthquake.usgs.gov/earthquakes/eventpage/us2000bqsn#moment-tensor>; gCMT, <http://www.globalcmt.org/CMTsearch.html>; IG-CEA, <http://www.csi.ac.cn/manage/eqDown/05LargeEQ/201711180634M6.9>

M_w (M_s)	Depth (km)	stk/dip/rak ($^{\circ}$)	Note
(6.9)	10.0	130/27/90	CENC
6.5	17.0	127/39/92	IG-CEA, Zhang et al. (2018)
6.3	19.2	124.7/59.0/88.8	IEF-CEA
6.4	8.0	132/55/95	USGS
6.5	12.0	119/24/72	gCMT
6.45	12.0	117.5/30.2/70.4	Wei et al. (2018)

surface thermal gradients have been observed (Craw et al., 2005). The teleseismic receiver function results also show the corresponding Moho uplift in the EHS (Zurek, 2008; Xu et al., 2013; Peng et al., 2017). The entire study area (Fig. 1) is divided into three subregions by the Indus-

Tsangpo Suture (ITS) and the Bangong-Nujiang Suture. Seismicity is mainly distributed along main sutures and faults, especially in the northern and eastern boundaries of the Namche Barwa massif, where the depths of most earthquakes are shallower than 30 km (Fig. 2). The location of the 2017 Mainling earthquake is at the northern side of the indentor corner, and the magnitude of this earthquake is the largest in the last 50 years.

Extensive geophysical studies have been performed in this area, including teleseismic P wave tomography (e.g., Ren and Shen, 2008; Li et al., 2008; Zhang et al., 2016; Liang et al., 2016; Peng et al., 2016), teleseismic receiver function migration and inversion (e.g., Zurek, 2008; Wang et al., 2010; Xu et al., 2013; Peng et al., 2017; Cheng et al., 2017), P_n tomography (e.g., Liang and Song, 2006; Zhou and Lei, 2016), earthquake surface wave dispersion tomography (e.g., Fu et al., 2010), joint inversion of surface waves and teleseismic body waves (Nunn et al., 2014), ambient noise and earthquake surface wave tomography (e.g., Yao et al., 2010), shear wave splitting analysis (e.g., Sol et al., 2007; Chen et al., 2015b; Chang et al., 2015), magnetotelluric imaging (e.g., Dong et al., 2016; Lin et al., 2017), and gravity modeling

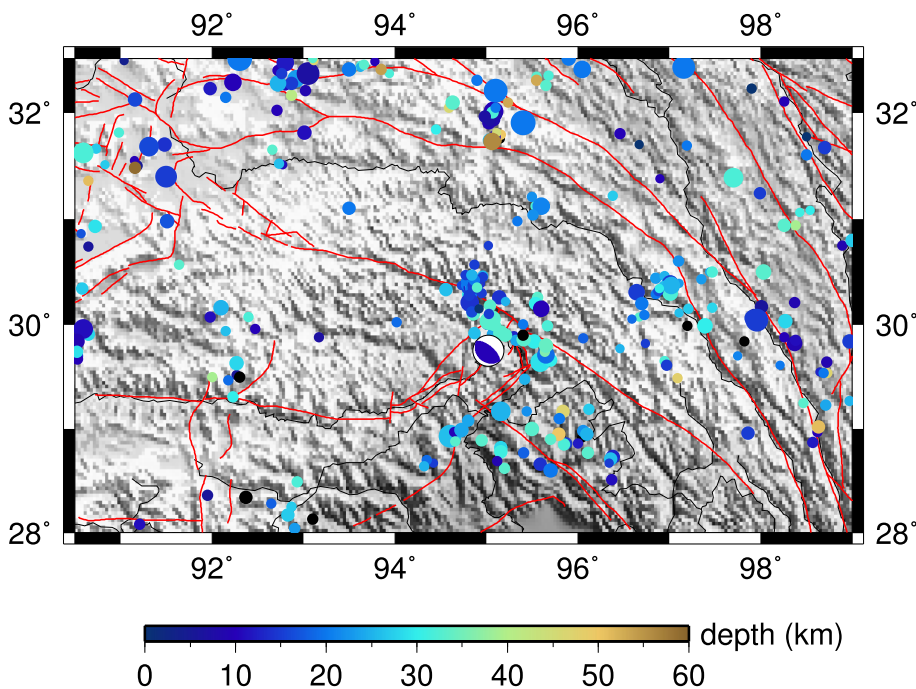


Fig. 2. Regional seismicity spatial distribution between 1970/03/20 and 2017/12/31. Depths of earthquakes are represented by colors. Magnitude of seismicity is represented by the size of circles. The beach ball denotes the location and focal mechanism of Mainling earthquake. Red lines denote major regional sutures and faults. Seismicity data are obtained from <http://www.csndmc.ac.cn/data/dir/formal/> (last access at 2018/03/30).

studies (e.g., Bai et al., 2013; Chen et al., 2015a). Most of these works focus on the relatively deep structures, but only the magnetotelluric imaging and receiver function studies have good resolution to resolve the velocity structures in the upper and middle crust, which are directly related to the nucleation mechanism of earthquakes.

In this study, by performing ambient noise tomography and considering the period-dependent ray tracing for periods of 5 ~ 40 s, we build a detailed 3-D crustal shear wave velocity model that has the highest resolution in the crust of the region to date. Our velocity model is compared with magnetotelluric imaging results and provides a detailed spatial distribution of the well-recognized mid-crustal low velocity zones (LVZs), which are usually explained as mechanically weak zones for the crustal channel flow in TP (e.g., Royden et al., 1997). Moreover, connection between the shear wave velocity model and the seismicity spatial distribution is revealed. The detailed shear wave velocity model, along with previous geological and geophysical observations, provides important constraints on the nucleation mechanism of the 2017 M_s 6.9 Mainling earthquake. In addition, together with seismicity distribution and previous metamorphism analysis (e.g., Koons et al., 2013), the characteristics of several vertical cross sections crossing the location of the Mainling earthquake are discussed. Finally, a tentatively integrated geodynamic model is proposed to reconcile the velocity model with previous observations.

2. Data and methods

Station distribution is shown in Fig. 1, which contains a 50-station array of broadband seismometers deployed during 2003–2004 as part of the Namche Barwa seismic experiment by Lehigh University (Sol et al., 2007), and a permanent station Lhasa (LSA), which is a part of the Global Seismographic Network. Empirical Green's functions are obtained by vertical component ambient noise cross-correlation, and inter-station Rayleigh wave phase velocities are measured using the image transformation method (e.g., Yao et al., 2008, 2006, 2010). In this study, we use inter-station phase velocities of $T = 5 \sim 40$ s instead of $T = 10 \sim 50$ s, which were used in a previous study (Yao et al., 2010). Yao et al. (2010) combined the dispersion data from ambient noise waveforms with teleseismic surface wave data, and their results have a lateral resolution of 150 ~ 200 km in the crust and upper mantle. In contrast, in this study, we focus on the high-resolution velocity structure in the crust, and only ambient noise data are utilized in the same study area. Fig. 3 shows the number of Rayleigh wave phase velocity measurements for each period, and an example of the raypath coverage at $T = 15$ s.

Direct surface wave tomography (Fang et al., 2015, DSurfTomo) is used to invert all of the measured Rayleigh wave phase traveltimes to obtain the 3-D crustal shear wave velocity model. In this study, only shear wave velocity is inverted for, whereas the P wave velocity and density are related to the shear wave velocity using an empirical relationship (Brocher, 2005). DSurfTomo uses the fast marching method (Rawlinson and Sambridge, 2004) to account for the ray bending effects of period-dependent raypaths, and achieves better tomographic reconstructions for complex media (Fang et al., 2015).

To determine the appropriate parameters in the inversion, we design a checkerboard test with a checkerboard size of $1^\circ \times 1^\circ$. Synthetic data are obtained using the fast marching method with a random Gaussian noise of 2%. Then, the synthetic data are inverted using the DSurfTomo algorithm (Fig. 4). The checkerboard test validates that the checkerboard pattern can be correctly recovered at depths of 5 ~ 50 km. Successful inversion also requires an appropriate initial model. Here, we use the surface wave dispersion inversion results of Shen et al. (2016) (after spatial smoothing) as the initial model, which uses Rayleigh wave phase and group velocity greater than 8 s and has a good resolution for the structures in the crust and upper mantle. Shen et al. (2016) obtained the 3-D shear wave velocity in a continental scale, but in this study, we aim to refine the velocity model in a smaller area with

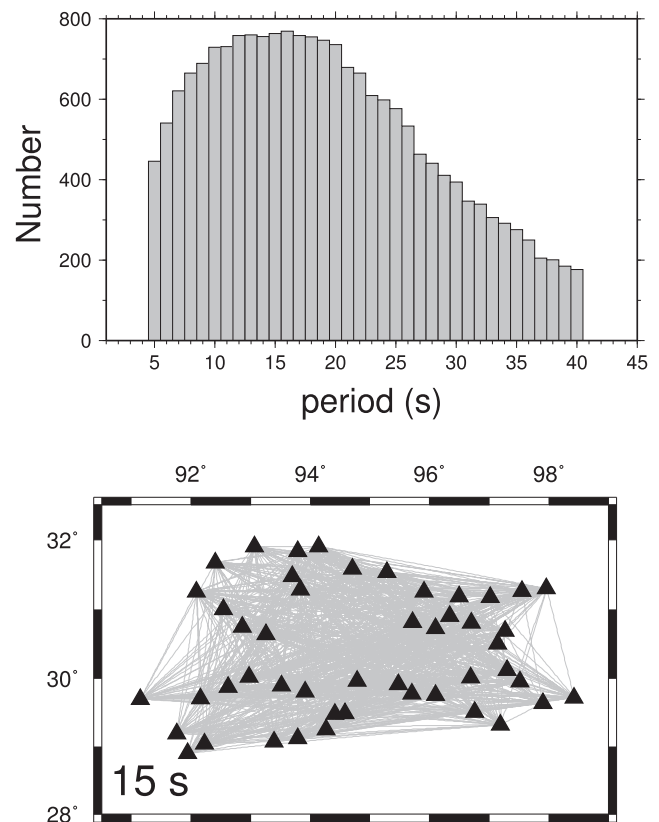


Fig. 3. Number of Rayleigh wave phase velocity measurements for each period used in the inversion and raypath coverage at a period of 15 s. Black triangles denote the station distributions.

higher resolution for the upper and middle crust (see supplementary Fig. S1 for comparisons between Shen et al. (2016)'s model and our refined velocity model). Fig. 5 shows the data fitting comparison between the initial model and the final model. The velocity structure at depths of 5 ~ 50 km is resolved by dispersion data, and the deeper part (> 50 km) of the initial model is conserved without inversion. To further validate the robustness of our inversion results, we also use the horizontally averaged velocity model from Shen et al. (2016)'s 3-D model as a 1-D initial model for inversion (Fig. S2). Tests show that the initial model has weak effects on the final inversion results. The corresponding inversion results are shown in Fig. S3.

3. Results

Using the parameters determined by the checkerboard tests, a 3-D shear wave velocity model is obtained using the DSurfTomo method (Figs. 6,7,8,9).

In the upper crust (Fig. 6), there is a relatively low velocity at the Namche Barwa massif that is surrounded by high velocity zones (HVZs) in the north, west, and east. A low velocity anomaly can be seen along 92°E , which is strikingly spatially consistent with the location of the north-south directional Comei rift (Fig. 6).

In the middle crust (Fig. 7), a dominant feature of the velocity model is a LVZ in the northwest and a HVZ in the southeast, which are separated by the ITS. The connectivity of the low and high velocity zones seems to vary with depth. The velocity of LVZs in middle crust is generally lower than 3.3 km/s. A clear LVZ can be seen along 92°E . Moreover, there is a roughly continuous HVZ along 97°E .

The cross sections along and orthogonal to the Jiali Fault (Fig. 8) show the distribution of high and low velocity anomalies changing with depths. Both cross sections exhibit a LVZ roughly above the location of the Mainling earthquake. Along the ITS, a mid-crustal LVZ exists

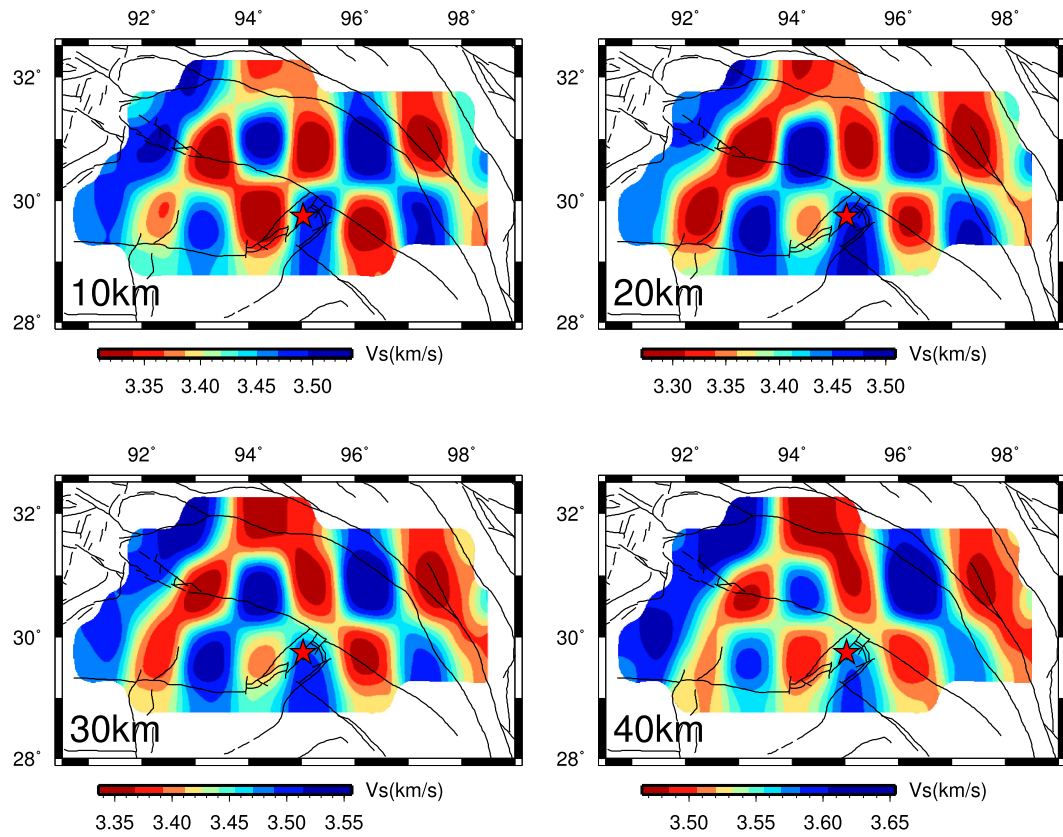


Fig. 4. Checkerboard test results for the inversion. The checkerboard grid size is $1^\circ \times 1^\circ$. The tests show that the checkerboard pattern can be successfully recovered at depths of 5–50 km.

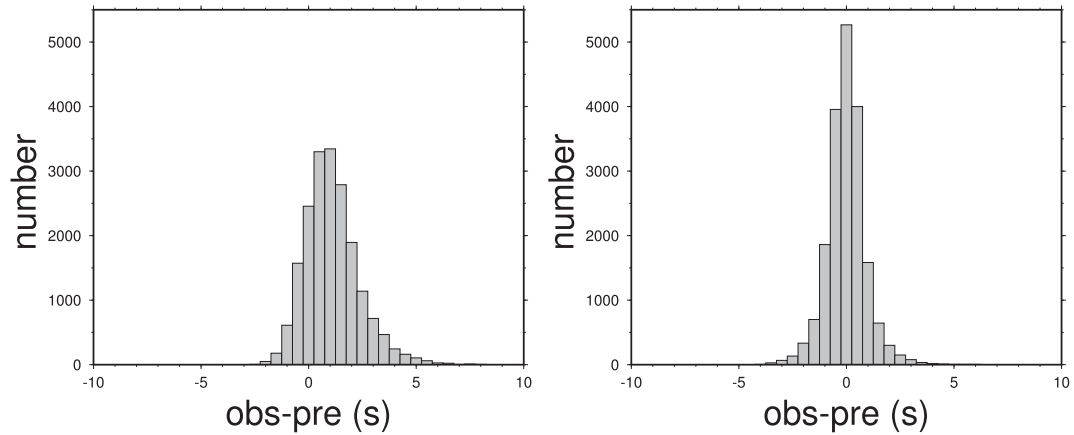


Fig. 5. Histograms to show the residual of data fitting with observations before (left) and after (right) inversion. The horizontal axis represents the difference between the observed and predicted travel time (observed travel time minus predicted travel time). The comparison shows that the final model fits the observations much better than the initial model does.

around 150 km in the northwest, but vertical to the ITS, there is a mid-crustal LVZ dipping from southwest to northeast. The cross section along 29.75°N (Fig. 9) shows a LVZ in the lower crust from approximately 150 km west to a slightly shallower location roughly beneath the Mainling earthquake. A HVZ can also be seen 150 km to the east. Moreover, vertical cross sections indicate that the seismicity distribution dips roughly from southwest to northeast (Figs. 8,9).

4. Discussion

4.1. Comparison with previous geophysical results

A comparison of our 3-D shear wave velocity model with magnetotelluric inversion results (Dong et al., 2016; Lin et al., 2017) shows that they are generally in agreement with each other in the middle crust (> 20 km), whereas they apparently disagree in the upper crust (< 20 km) (Figs. 6,7). This result is not surprising because the magnetotelluric data are very sensitive to water content, whereas the water content in the upper crust has a weak effect on the shear wave velocity. The velocity in the upper crust is related mainly to the distribution of

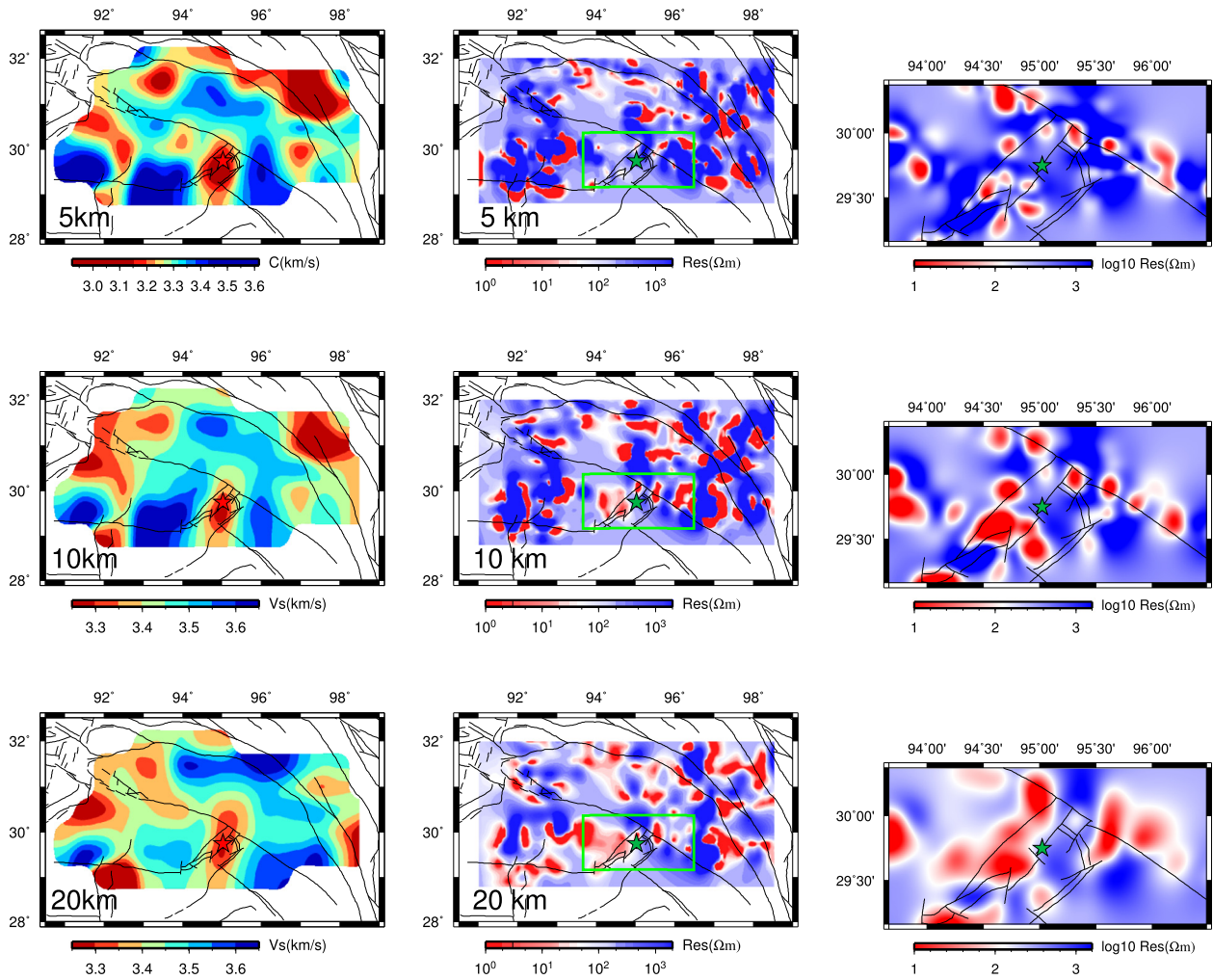


Fig. 6. Horizontal slices of the 3-D velocity model for the upper crust and corresponding magnetotelluric inversion results of Dong et al. (2016) (middle) and Lin et al. (2017) (right) at 5 km, 10 km and 20 km. Dong et al. (2016)'s results are clipped to be comparable to our velocity model. The green rectangles in the middle figures denote the ranges of the corresponding figures on the right.

sedimentary basins and mechanically weak zones (e.g., Yang et al., 2012).

Since we invert for shear wave velocity by using Rayleigh wave dispersion data, which are sensitive to the vertically polarized shear wave speed (V_{sv}), the LVZs in the middle crust of the velocity model can be attributed to either anisotropic minerals aligned by deformation or partial melts (e.g., Yang et al., 2012). By comparing the spatial distribution of V_{sv} and the radial anisotropy results of Shapiro et al. (2004), Yang et al. (2012) concluded that, particularly for central and western Tibet, the mid-crustal LVZs are at least partially due to the mineralogical alignment, which may result from past deformation. In contrast, for the area surrounding the EHS in this study, the general agreement of spatial distribution of LVZs and the high conductivity zones is in favor of the model suggesting that the middle crust is currently mechanically weak, probably due to presence of partial melts, and has the potential for mid-crustal channel flow (e.g., Royden et al., 1997; Beaumont et al., 2001). The explanation is further supported by the geothermal observations (Hu et al., 2000) and textural analysis with phase equilibrium modeling (Guilmette et al., 2011). In addition, both our velocity model and the magnetotelluric inversion results show that some of the LVZs and HVZs are spatially correlated to major faults (Yao et al., 2010), e.g., the obvious boundary around the ITS between the high and low velocity zones on the western boundary of the EHS at a depth of 30 km.

Previous studies have shown that the mid-crustal conductive

anomaly caused solely by aqueous fluid is not possible (e.g., Hu et al., 2000; Dong et al., 2016). Therefore, the velocity and resistivity anomalies in the middle crust are mainly due to the combination of the water content, higher temperature, and chemical composition. Dong et al. (2016) estimated in detail the fluid fraction based on the magnetotelluric inversion result and found that the NNE-SSW striking low fluid/melts fraction band around 97°E of the entire crust truncates the conductive middle to lower crust and the upper mantle into two separate regions (Figs. 6, 7, middle panel). Dong et al. (2016)'s results suggest the disconnectivity of the crustal channel flow in the E-W direction and support their 'extensional extrusion' geodynamic model around the EHS (Dong et al., 2016). Our velocity model generally agrees with their magnetotelluric imaging results and has a HVZ in the middle crust around 97°E (Figs. 6, 7, left panel). In addition, our model reveals that the distribution of the LVZs follows a complex 3-D pattern and that the connectivity of the LVZs varies with depth. Previous studies have indicated that the existence of partial melts is plausible with a shear wave velocity of 3.3 km/s or less at a depth of 30 km (Yang et al., 2012). Since the existence of aqueous fluid cannot be excluded, the LVZs can be explained by the joint effects of partial melt and aqueous fluid.

Lin et al. (2017) obtained a high-resolution 3-D resistivity map around the EHS with denser MT stations in a smaller region (Figs. 6, 7, right panel). It seems that our velocity model agrees with Lin et al. (2017)'s conductivity model better than it agrees with Dong et al.

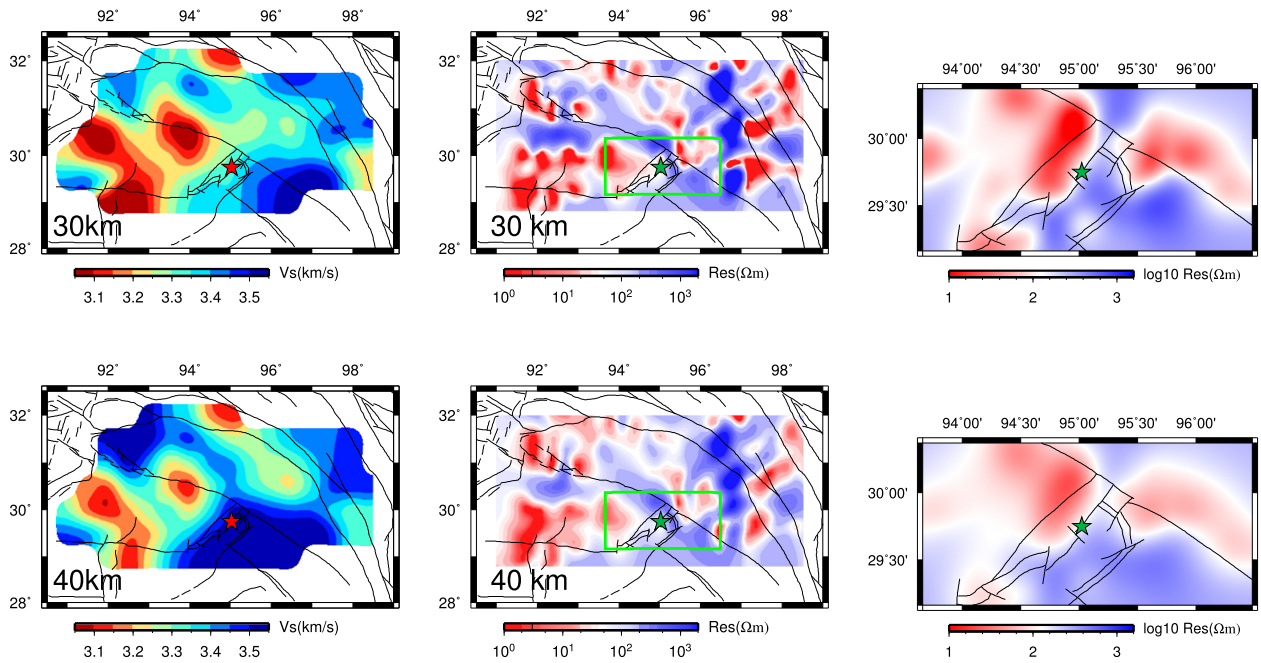


Fig. 7. Horizontal slices of the 3-D velocity model for the middle crust and corresponding magnetotelluric inversion results of Dong et al. (2016) (middle) and Lin et al. (2017) (right) at 30 km and 40 km. For the symbols' meanings, see the caption of Fig. 6 for details.

(2016)'s model around the EHS within this smaller region in the middle crust (Fig. 7). Lin et al. (2017) found a resistive upper crust at shallow depths of the EHS, and they interpreted it as brittle rocks or a Mesozoic overthrust arc, which can be observed as the LVZ in the upper crust in our velocity model. Another robust observation shared by our velocity model and previous magnetotelluric imaging results (Dong et al., 2016; Lin et al., 2017) is the LVZ along rift at 92°E, which is also revealed by previous studies (e.g., Ren and Shen, 2008; Fu et al., 2010) down to a

deeper part of the upper mantle at approximately 300 km.

4.2. Structural control of the Mainling earthquake and regional seismicity

In the horizontal slices (Fig. 6), it is clear that the horizontal location of the 2017 M_s 6.9 Mainling earthquake lies where the velocity gradient is the largest at the depth ~ 30 km of the shear wave velocity model. Moreover, the seismicity is mostly distributed on the northern

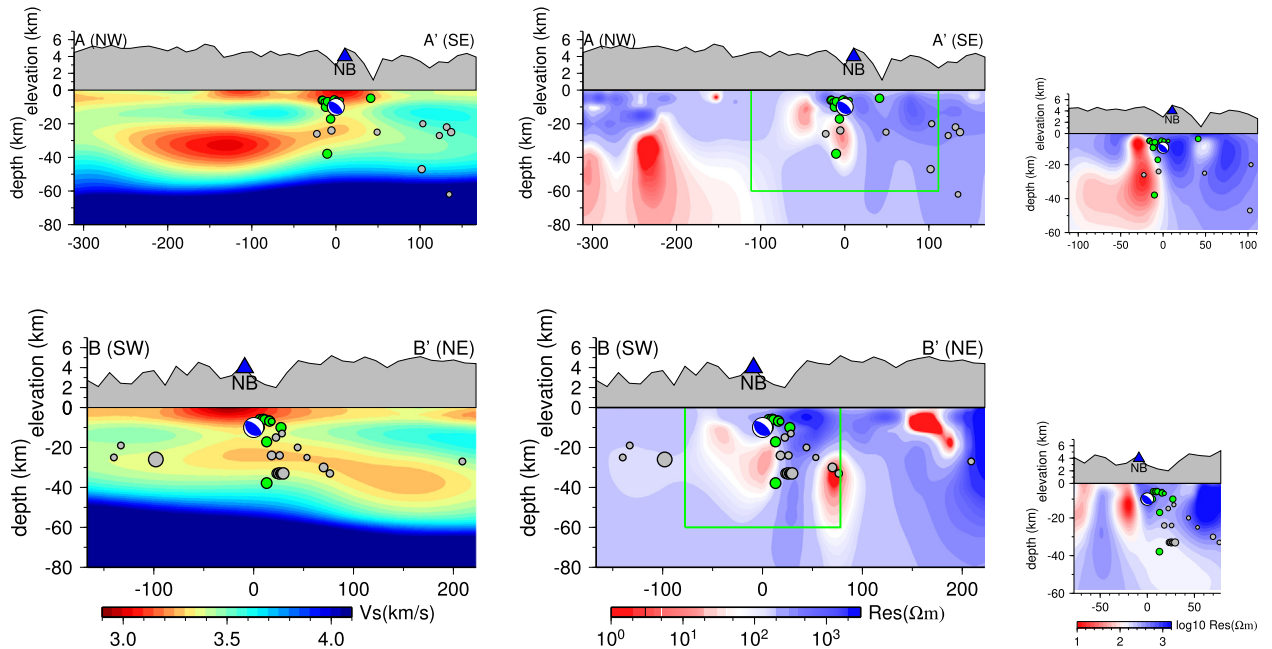


Fig. 8. AA' (up) and BB' (down) Vertical slices of the 3-D velocity model and corresponding magnetotelluric inversion results of Dong et al. (2016) (middle) and Lin et al. (2017) (right). The green rectangles in the middle figures denote the ranges of the corresponding figures on the right. The blue beach ball denotes the focal mechanism of the Mainling earthquake. Circles denote seismicity ($M_w > 3.0$) within a range of 20 km from the cross sections between 1970/02/08 and 2017/12/31 (seismicity data obtained from <http://www.csndmc.ac.cn/data/dir/formal/>, last access on 2018/03/20). Gray and green circles denote the seismicity before and after the Mainling earthquake, respectively. The sizes of the circles denote the earthquake magnitudes. The blue triangle represents the location of Namche Barwa peak (NB). For details of the location of the cross sections, see Fig. 1.

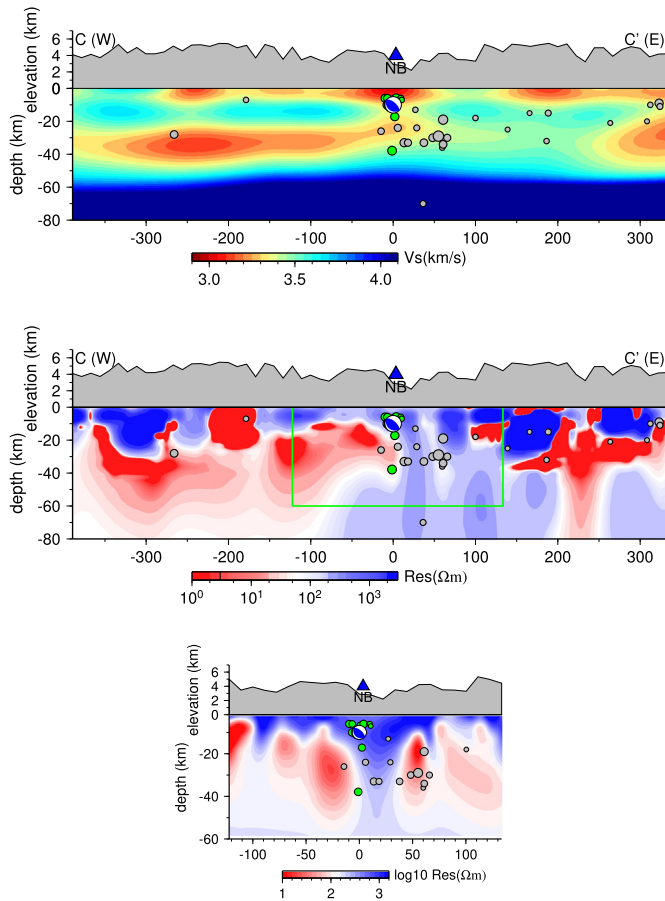


Fig. 9. CC' vertical cross section along 29.75°N, which crosses the location of the Mainling earthquake. For the symbols' meanings, see the caption of Fig. 8 for details.

and eastern boundaries of the EHS (Fig. 2), which also corresponds to the horizontal large velocity gradient at 30 km. Cross sections of the velocity model (Figs. 8,9) show that the seismicity distribution follows the dipping geometry of the local faults and the northeastern subduction direction of the Indian plate beneath the Eurasian plate. In addition, the depths of most of the earthquakes are shallower than 30 km, indicating the ductile characteristics of the middle crust in this area (e.g., Yang et al., 2012; Dong et al., 2016).

The densely distributed seismicity demonstrates that the northern and eastern parts of the EHS are active and that the local stress is intensive, which is supported by recent GPS observations (e.g., Gupta et al., 2015; Zheng et al., 2017) and a geochronology analysis (e.g., Xu et al., 2012). The spatial relationship between the distribution of earthquakes and the shear wave velocity model implies a possible seismicity nucleation mechanism as follows: the relatively rigid upper crust in the northwest, along with the relatively weak middle crust, moves southeastward, but its movement is blocked by the more rigid middle crust in the east, where the local stress in the upper crust is accumulated and dense seismicity is induced. It is noteworthy that the main stress accumulation in this area is in the N-S direction and results from the collision of the Indian and Eurasian plates. The additional stress accumulation mechanism discussed above is consistent with the generally E-W extensional direction in TP. The nucleation mechanism of the 2017 Mainling earthquake is a combination of the stress accumulation effects, major fault direction, and the aqueous fluid and geothermal states (as discussed below). The focal mechanism of the Mainling earthquake was thrust-type, and it shows a NE-SW compressional direction, which is consistent with the direction of continental subduction. Further, the aftershock distribution in NW-SE (Wei et al.,

2018; Yin et al., 2018) and the rupture direction of the Mainling earthquake coincide with the additional stress accumulation direction.

Seismicity distribution is also closely related to the aqueous fluid and geothermal states. Wang et al. (2009) analyzed the triggering mechanism of the M_w 7.9, 2008 Wenchuan earthquake using their tomography result, and they proposed a mechanism in which the upward migration of deep-seated aqueous fluid triggered the earthquake by reducing the mechanical strength of the brittle layer. This mechanism can also be used to explain the triggering mechanism of Mainling earthquake, which is further supported by the local observations that the LVZs seem to go upward beneath the Namche Barwa massif (Figs. 9) and upward of the Moho's depth (e.g., Xu et al., 2013), geotherm (Craw et al., 2005), and conductivity (Lin et al., 2017). In addition, on a smaller scale, by analyzing the repeated earthquake swarm sequence along the Jiali Fault in the EHS, Mukhopadhyay and Dasgupta (2015) proposed that the hydro fracturing may have generated the swarms, in terms of pore pressure variation. Their results also indicate the importance of aqueous fluid to the nucleation of earthquakes.

4.3. Implications for regional tectonic models

Although the resolution of our velocity model is limited to the depth of the middle to lower crust (~ 50 km depth), the deep structures in the upper mantle strongly influence the shallower tectonic evolution. Recently, accumulating geophysical evidence has indicated that the subduction dipping geometry of the Indian plate into the Eurasian plate varies in the E-W direction (e.g., Nunn et al., 2014; Chen et al., 2015b; Liang et al., 2016) and that slab tears and mantle material upwells into the crust beneath the EHS (Zhang et al., 2016; Xu et al., 2013; Chen et al., 2015b; Tian et al., 2015; Liang et al., 2016; Peng et al., 2016). A gravity anomaly study showed an Indian mantle delamination in eastern Tibet from 87°E to 93°E (Chen et al., 2015a). This delamination was also proposed by Xu et al. (2012) in their two-phase model to explain the formation of the EHS. The LVZs, explained as an upwelling path of mantle material, can be found in the regional tomographic models of Ren and Shen (2008) and Fu et al. (2010), in which the LVZ extends to a depth of approximately 300 km. In our velocity model, the upwelling path can also be seen along 92°E in the upper and middle crust, which correlates to the location of the N-S-oriented Comei rift zone (Fig. 1).

Currently, two types of models exist to explain the mechanisms underlying the deformation and expansion of the TP: crust-mantle decoupling due to lower crust channel flow (e.g., Royden et al., 1997, 2008) and the vertically coherent deformation model (e.g., Tapponnier et al., 1982; Sol et al., 2007; Wang et al., 2008). To reconcile the contradiction of these two models, Fu et al. (2010) proposed that the crust and mantle deform coherently for strong parts and that channel flow exists in the weaker parts, but the possibility of this requires further validation via geodynamic simulation. Yao et al. (2010) found that the level of decoupling of the upper crust and the lithospheric mantle may vary laterally by analyzing the surface wave azimuthal anisotropy. Dong et al. (2016) proposed an 'extrusion process' model that can explain the observation of GPS direction and their magnetotelluric inversion results. Significant evidence of the 'extrusion process' model around the EHS is the high resistivity zone that truncates the whole crust and the upper mantle along 97°E, and this can be seen as a HVZ in our shear wave velocity model (Fig. 7).

Further, the 'tectonic aneurysm' model was proposed to explain the formation of the upper crust of the EHS, which emphasizes the coupling between surface process and tectonics and has predicted localized exhumation and rock uplift in the Namche Barwa massif since 5 Ma (Zeitler, 2001; Koons et al., 2013). The coupling effects between atmospheric effects and tectonics have been believed to be an important factor in exciting and maintaining extreme orogens, where exhumation is both rapid and localized (Bendick and Ehlers, 2014; Lin et al., 2017). The existence of coupling effects is supported by geomorphology

studies (e.g., Finnegan et al., 2008; Enkelmann et al., 2011), a thermal structure study (Craw et al., 2005), and the discovery of the deeply incised canyon of the Yarlung Tsangpo River (Wang et al., 2014). In our velocity model, one of the most interesting features in the upper crust is the relatively low velocity around the EHS, which is bounded by two high velocity anomalies in the west and east (Fig. 6). In the vertical cross sections (Fig. 8), a LVZ can be clearly seen in the upper crust, immediately above the location of the Mainling earthquake. The ‘tectonic aneurysm’ model contains several main aspects, including that the local rocks are weakened as the strong upper crust is stripped away by erosion, the local elevation is maintained by crustal material flow, and the geotherm is elevated by the rapid uplift of hot rock (Burchfiel and Chen, 2012; Koons et al., 2013). The complete process results in a weak upper crust, which corresponds to the relatively low velocity in our shear wave velocity model. Lin et al. (2017) observed a local high resistive upper crust, which they interpreted as brittle rocks or a Mesozoic overthrust arc. Since the rocks in the Namche Barwa massif are continuous with the High Himalayan crystalline zone in the Himalaya (Burchfiel and Chen, 2012), which are different from surrounding areas, the properties of the rocks also contribute to the relatively low velocity anomaly in the upper crust.

To reconcile our velocity model with previous geophysical and geological observations from the upper mantle to the surface in this region, here, we propose a tentatively integrated geodynamic model to explain the local stress condition and seismicity distribution around the EHS, the validation of which relies on more refined velocity models and confirmation by other kinds of datasets.

- Since the Indian plate underthrust beneath the Eurasian plate, an indenter formed at the corner of southeastern Tibet. The rock of indenter is highly metamorphism and continuous with the High Himalayan crystalline zone (Burchfiel and Chen, 2012). The indenter squeezed into the Lhasa block, forming the antiform EHS. The EHS continuously moved northward as the Indian plate moved forward (Whipple, 2014; Yang et al., 2018).
- Due to the tearing of the Indian subduction slab beneath the Eurasian plate, a upwelling path was formed for mantle material going up to the bottom of Moho. The mantle material acted as a heating source around 92°E. Because of the subduction, the regional crust was thickened, and both the local compressional stress in the N-S direction and extensional stress in the E-W direction were formed around the EHS. Further, the extensional stress in the E-W direction created weakened passages for hydrous fluids (Dong et al., 2016). Due to the effects of the heating source and thickened crust, a regional crustal ‘channel flow’ may have been formed by the joint effect of partial melts of crustal material and existence of aqueous fluid along the weakened passages (Fig. 7).
- The rock uplift resulted in a relatively weak zone beneath the EHS antiform. The ‘channel flow’ tended to flow into the weak zone beneath the EHS. Moreover, a high erosion rate was caused by high local elevation. 3-D geometry subduction, together with the high erosion rate of the EHS, excited extreme orogens where exhumation was both rapid and localized; this process created feedback between exhumations and uplift (e.g., Bendick and Ehlers, 2014; Koons et al., 2013). The ‘tectonic aneurysms’ model can be used to describe the feedback process, which contains the weakening of local rocks and material flowing into the local weak zones, and the formation of both the upward of the ‘channel flow’ beneath the Namche Barwa massif (Fig. 9) and the weak zones in the upper crust (Fig. 6). In addition, the geometry of ‘channel flow’, together with the content of fluid and the local stress condition, can be used to explain the local seismicity distribution and nucleation mechanism of the Mainling earthquake.
- Along the subduction geometry (Fig. 8), the distribution of LVZs can be explained by the mechanism proposed by the numerical study of Beaumont et al. (2001), who emphasized the coupling effects of low-

viscosity crustal channel and surface denudation.

5. Conclusions

In this study, we determine the 3-D detailed crustal shear wave velocity in and around the eastern Himalayan syntaxis by using direct ambient noise tomography with Rayleigh wave phase velocity data. Our velocity model is extensively compared with previous magnetotelluric inversion results and reveals the distribution and connectivity of the LVZs. The LVZs seem to follow complex 3-D patterns and are spatially correlated to major faults. Moreover, we discuss the nucleation mechanism of the 2017 M_s 6.9 Mainling earthquake and the local seismicity distribution. The seismicity nucleation is attributed to the local stress condition and velocity structure and to the aqueous fluid and geothermal states. In addition, our velocity model exhibits a low velocity anomaly in the upper crust around the Namche Barwa massif, which indicates a local weak zone. The local weak zone can be explained by the conceptional ‘tectonic aneurysm’ model, which emphasizes the coupling effects of climate, erosion and tectonics. Finally, we propose a tentatively integrated geodynamic model to explain both our velocity model and previous observations.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.tecto.2018.06.006>.

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